

## C.1 Proof of the validity of the IW estimator

### Proof of Proposition 5

*Proof.* Provided that the DID estimator is well-defined with pre-period  $s$  and control cohorts  $C$ , we first show the DID estimator is an unbiased and consistent estimator for  $E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e] - E[Y_{i,e+\ell} - Y_{i,s} \mid E_i \in C]$ . We prove the first term in the DID estimator  $\frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]}$  is unbiased and consistent for  $E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e]$ . The argument for the second term in the DID estimator follows similarly.

For unbiasedness, note that by the Law of Iterated Expectations and linearity of  $\mathbb{E}_N$ , we have

$$E\left[\frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]}\right] = E\left[E\left[\frac{\mathbb{E}_N[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]} \mid E_i\right]\right] \quad (74)$$

$$= E\left[\frac{\mathbb{E}_N\left[E[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\} \mid E_i]\right]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]}\right] \quad (75)$$

$$= E\left[\frac{\mathbb{E}_N\left[E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e] \cdot \mathbf{1}\{E_i = e\}\right]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]}\right] \quad (76)$$

$$= E\left[\frac{E[(Y_{i,e+\ell} - Y_{i,s}) \mid E_i = e] \cdot \mathbb{E}_N[\mathbf{1}\{E_i = e\}]}{\mathbb{E}_N[\mathbf{1}\{E_i = e\}]}\right] \quad (77)$$

$$= E[(Y_{i,e+\ell} - Y_{i,s}) \mid E_i = e]. \quad (78)$$

For consistency, by the Law of Large Numbers the numerator and the denominator converge in probability to  $E[(Y_{i,e+\ell} - Y_{i,s}) \cdot \mathbf{1}\{E_i = e\}]$  and  $Pr\{E_i = e\}$  respectively. By the Law of Iterated Expectations and Slutsky's theorem, it converges in probability to  $E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e]$ .

To show that the DID estimator is an unbiased and consistent estimator for  $CATT_{e,\ell}$ , it remains to show  $E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e] - E[Y_{i,e+\ell} - Y_{i,s} \mid E_i \in C] = CATT_{e,\ell}$ .

Since  $s < e$  and  $c > e + \ell$ , we have

$$E[Y_{i,e+\ell} - Y_{i,s} \mid E_i = e] - E[Y_{i,e+\ell} - Y_{i,s} \mid E_i \in C] \quad (79)$$

$$= E[Y_{i,e+\ell}^e - Y_{i,s}^e \mid E_i = e] - \sum_{c \in C} Pr\{E_i = c \mid E_i \in C\} E[Y_{i,e+\ell}^c - Y_{i,s}^c \mid E_i = c] \quad (80)$$

$$= E[Y_{i,e+\ell}^e - Y_{i,s}^\infty \mid E_i = e] - \sum_{c \in C} Pr\{E_i = c \mid E_i \in C\} E[Y_{i,e+\ell}^\infty - Y_{i,s}^\infty \mid E_i = c] \quad (81)$$

$$= E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty \mid E_i = e] + E[Y_{i,e+\ell}^\infty - Y_{i,s}^\infty \mid E_i = e] \quad (82)$$

$$- \sum_{c \in C} Pr\{E_i = c \mid E_i \in C\} E[Y_{i,e+\ell}^\infty - Y_{i,s}^\infty \mid E_i = c] \quad (83)$$

$$= E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty \mid E_i = e] + E[Y_{i,e+\ell}^\infty - Y_{i,s}^\infty] - E[Y_{i,e+\ell}^\infty - Y_{i,s}^\infty] \quad (84)$$

$$= E[Y_{i,e+\ell}^e - Y_{i,e+\ell}^\infty \mid E_i = e] \quad (85)$$

where the second equality follows from Assumption 2 and the fourth equality follows from Assumption 1.  $\square$

## Proof of Proposition 6

*Proof.* We first show the consistency and asymptotic normality of the weight estimators. The weight estimators  $\hat{\mathbf{f}}^\ell$  are consistent since  $\frac{N_e}{\sum_{e \in h^\ell} N_e} \rightarrow_p \frac{Pr\{E_i = e\}}{Pr\{E_i \in h^\ell\}} = Pr\{E_i = e \mid E_i \in h^\ell\}$  by the Law of Large Numbers and Slutsky's theorem. Note that  $\frac{N_e}{\sum_{e \in h^\ell} N_e}$  is also the regression coefficient estimator from the following cross-sectional regression

$$\mathbf{1}\{E_i = e\} = \beta \mathbf{1}\{E_i \in h^\ell\} + \eta_i(e) \quad (86)$$

with population regression coefficient equal to  $\beta = Pr\{E_i = e \mid E_i \in h^\ell\}$ . Then by OLS asymptotics which holds as  $E_i$  are iid by assumption and  $\eta_i(e)$  is bounded, we have

$$\sqrt{N} \left( \frac{N_e}{\sum_{e \in h^\ell} N_e} - Pr\{E_i = e \mid E_i \in h^\ell\} \right) \rightarrow_d N \left( 0, \frac{E[\mathbf{1}\{E_i \in h^\ell\}^2 \eta_i^2(e)]}{E[\mathbf{1}\{E_i \in h^\ell\}]^2} \right). \quad (87)$$

Note that  $\mathbf{1}\{E_i \in h^\ell\}^2 = \mathbf{1}\{E_i \in h^\ell\}$  so the asymptotic variance is equal to

$$\frac{E[\eta_i^2(e) \mid E_i \in h^\ell] Pr\{E_i \in h^\ell\}}{Pr\{E_i \in h^\ell\}^2} = \frac{E[\eta_i^2(e) \mid E_i \in h^\ell]}{Pr\{E_i \in h^\ell\}}. \quad (88)$$

Similarly, for a pair of cohorts with  $e \neq e'$ ,  $\frac{N_e}{\sum_{e \in h^\ell} N_e}$  and  $\frac{N_{e'}}{\sum_{e' \in h^\ell} N_{e'}}$  are asymptotically correlated with covariance  $E[\eta_i(e) \eta_i(e') \mid E_i \in h^\ell] / Pr\{E_i \in h^\ell\}$ . Thus,  $vec(\hat{\mathbf{f}}^\ell)$  has asymptotic distribution

$$\sqrt{N} \left( vec(\hat{\mathbf{f}}^\ell) - vec(\mathbf{f}^\ell) \right) \rightarrow_d N \left( 0, \Sigma_{f^\ell} \right). \quad (89)$$

Here  $\Sigma_{f^\ell}$  is a matrix with diagonal entries equal to  $\frac{E[\eta_i^2(e) \mid E_i \in h^\ell]}{Pr\{E_i \in h^\ell\}}$ , and off-diagonal entries equal to  $\frac{E[\eta_i(e) \eta_i(e') \mid E_i \in h^\ell]}{Pr\{E_i \in h^\ell\}}$ .

By consistency of the weight estimators and Proposition 5, we prove the consistency of the IW estimator,

the first part of Proposition 6.

We next show the asymptotic normality of the  $\widehat{\delta}_{e,\ell}$ . The standard OLS asymptotics applies because by assumption after double demeaning, the data  $(\ddot{\mathbf{B}}_{i,t}, \ddot{\epsilon}_{i,t})$  is iid across  $i$  and has nonzero finite fourth moments.

The asymptotic distribution of this estimator is thus

$$\sqrt{N}(\widehat{\delta} - \delta) \rightarrow_d N\left(0, V_{\ddot{\mathbf{B}}}^{-1} \text{Var}\left(\sum_{t=0}^{T-1} \ddot{\mathbf{B}}_{i,t} \ddot{\epsilon}_{i,t}\right) V_{\ddot{\mathbf{B}}}^{-1}\right) \quad (90)$$

where  $V_{\ddot{\mathbf{B}}} = \sum_{t=0}^{T-1} E[\ddot{\mathbf{B}}_{i,t} \ddot{\mathbf{B}}_{i,t}^\top]$ .

Lastly, by the delta method, we have

$$\sqrt{N}\left(\text{vec}(\widehat{\mathbf{f}}^\ell)^\top \widehat{\delta} - \text{vec}(\mathbf{f}^\ell)^\top \delta\right) \rightarrow_d N\left(0, \delta^\top \Sigma_{f^\ell} \delta + \Sigma_\ell\right) \quad (91)$$

where  $\Sigma_\ell = \text{vec}(\mathbf{f}^\ell)^\top V_{\ddot{\mathbf{B}}}^{-1} \text{Var}\left(\sum_{t=0}^{T-1} \ddot{\mathbf{B}}_{i,t} \ddot{\epsilon}_{i,t}^2 \ddot{\mathbf{B}}_{i,t}^\top\right) V_{\ddot{\mathbf{B}}}^{-1} \text{vec}(\mathbf{f}^\ell)$ . This follows because  $\text{vec}(\widehat{\mathbf{f}}^\ell)$  and  $\widehat{\delta}$  are uncorrelated: the asymptotic covariance between  $\frac{N_e}{\sum_{e \in h^\ell} N_e}$  and  $\widehat{\delta}$  is equal to

$$\frac{V_{\ddot{\mathbf{B}}}^{-1} \text{Cov}(\mathbf{1}\{E_i \in h^\ell\} \eta_i(e), \ddot{\mathbf{B}}_{i,t} \ddot{\epsilon}_{i,t})}{E[\mathbf{1}\{E_i \in h^\ell\}^2]^2} = \frac{V_{\ddot{\mathbf{B}}}^{-1} E[\ddot{\mathbf{B}} \eta_i(e) \ddot{\epsilon}_{i,t} \mid E_i \in h^\ell]}{E[\mathbf{1}\{E_i \in h^\ell\}^2]^2}. \quad (92)$$

Since  $\ddot{\mathbf{B}}$  and  $\eta_i(e)$  are functions of  $E_i$ , we have  $E[\ddot{\mathbf{B}} \eta_i(e) \ddot{\epsilon}_{i,t} \mid E_i \in h^\ell] = E[\ddot{\mathbf{B}} \eta_i(e) E[\ddot{\epsilon}_{i,t} \mid E_i, E_i \in h^\ell]]$ .

Furthermore, specification (26) is saturated in  $E_i$  and relative time so  $E[\ddot{\epsilon}_{i,t} \mid E_i] = 0$ . This proves the

asymptotic normality of the IW estimators, the second part of Proposition 6.  $\square$